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Bachelor of Computer Science (Hons)
Bachelor of Software Engineering (Hons)

ASSIGNMENT COVER SHEET

Module Code: **ITS64304**

Module Name: **Theory of Computation**

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Assignment No. / Title	Assignment 3
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Declaration *(need to be signed by students. Otherwise, assignment will not be marked)*

We certify that this assignment is entirely our own work, except where we have given fully documented references to the work of others, and that the material contained in this assignment has not previously been submitted for assessment in any other formal course of study.

Signature of Students: xy , txy , sy , le , hc , syao

Marks: 17/20	**Evaluated by:** Jafhate Edward Digitally signed by Jafhate Edward Date: 2025.07.23 11:18:35 +08'00'

Evaluator's Comments:

Q1-Q4 were exceptional. while all CAs were listed correctly.However, Q5 needs to be elaborated more and explain in-depth with conciseness. Too vague. Key points could have been explained more

Question 1

Suppose your RSA modulus is n = 187 and your encryption key e = 7, Find the decryption key d. Show your working. Is this possible for large values of n say of the order of 1024 bits? Give your reasons in one or two sentences.

Answer:

Step 1: Factor n into p and q

$n = p \times q$

$187 = 17 \times 11$

$p = 17, q = 11$

Step 2: Compute $\phi(n)$

$\phi(n) = (p - 1)(q - 1)$

$= (17 - 11) \times (11 - 1)$

$= 16 \times 10$

$= 160$

Step 3: Find the Decryption Key d

$(d \times e) \, mod \, \phi(n) = 1$

$(d \times 7) \, mod \, 160 = 1$

$(d \times 7) = 1 \, mod \, 160$

$161 \, mod \, 160 = 1$

$\therefore d = 23$

By using Euclidem's algorithm

$d7 \, mod \, 160 = 1$

$160 = n(7) + remainder$

$= 22(7) + 6$

$7 = n(6) + remainder$

$= 1(6) + 1$

Equation 1: $1 = 7 - 1(6)$

Equation 2: $6 = 160 - 22(7)$

Sub E2 into E1: $1 = 7 - 1(160 - 22(7))$

$= 1(7) - 1(160) + 22(7)$

$= 23(7) - 1(160)$

$\therefore d = 23$

Is this possible for large values of n say of the order of 1024 bits?

Finding the RSA decryption key d is impractical for large values of n (like 1024-bit numbers) because factoring such large numbers into primes is extremely difficult, even with powerful algorithms like the General Number Field Sieve(Case, n.d.).

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Question 2

Suppose your RSA modulus is n = 119 and your decryption is d = 77, Find the encryption key e. Show your working. Is this possible for large values of n say of the order of 1024 bits? Give your reasons in one or two lines.

Answer:

Step 1: Factor n into p and q

$n = p \times q$

$119 = 7 \times 17$

$p = 7, q = 17$

Step 2: Compute $\phi(n)$

$\phi(n) = (p - 1)(q - 1)$

$= 6 \times 16$

$= 96$

Step 3: Use the modular inverse condition to find the encryption key, e

$(d \times e) \, mod \, \phi(n) = 1$

$(77 \times e) \, mod \, 96 = 1$

$(77 \times e) = 1 \, mod \, 96$

$385 \, mod \, 96 = 1$

$\therefore e = 5$

By using Euclidem's algorithm

$77e \, mod \, 96 = 1$

$96 = n(77) \quad + remainder$

$= 1(77) + 19$

$77 = n(19) + remainder$

$= 4(19) + 1$

Equation 1: $1 = 77 - 4(19)$

Equation 2: $19 = 96 - 1(77)$

Sub E2 into E1: $1 = 77 - 4(96 - 1(77))$

$= 1(77) - 4(96) + 4(77)$

$= 5(77) - 4(96)$

$\therefore e = 5$

Is this possible for large values of n say of the order of 1024 bits?

Direct computation of the encryption key from the decryption key in RSA is not feasible for large n values such as 1024 bits because it requires factoring n into its prime factors, which is impractical with current technology. This difficulty in factoring large n values is what keeps RSA secure.

2

Question 3

Alice wishes to invest in shares in MoMo a food chain business in Malaysia. Alice sends an authenticated message M = 8 to his professional fund manager Bob using RSA algorithm where his public key n = 143, e = 103 and private key d = 7. Derive the signature that would have been used by Alice. Explain how Bob would verify Alice's signature.

Answer:

Step 1: Signature Generation by Alice

Alice uses her private key, d = 7 to sign the authenticated message, M = 8 and given that n = 143. The signature, S would be:

$S = M^d \bmod n$

$= 8^7 \bmod 143$

$= 57$ ✓

Step 2: Verification by Bob

Bob receives the authenticated message, M = 8 along with the signature S = 57. He then uses Alice's public key, e = 103 and n = 143 to verify whether the signature, S = 47 corresponds to the authenticated message, M = 8.

$S^e \bmod n$

$= 57^{103} \bmod 143$

$= 8$

$= M$ ✓

Alice uses her private key d=7 to sign the message. Bob uses Alice's public key e=103 to verify Alice's signature and get back the original message as 8. As Alice's public key and private key are a pair and mathematically related, nobody else other than Alice would have signed the message.

Since Alice's public key, e can successfully verify that the signature, S generated using her private key, d corresponds to the authenticated message, M, this confirms that the signature is valid. It also proves that the message was indeed sent by Alice and has not been altered.

✓

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Question 4

Consider the two primes p and q in RSA algorithm are taking a value 11 and 13, respectively.

Derive the encrypted text for the plain text JEFFREY ULLMAN and then decrypt the original plain text. Show the encryption and decryption process/steps in terms of a simple

table. The ASCII values for the uppercase English alphabets is given below.

Table: ASCII values

Name	Hex
A	41
B	42
C	43
D	44
E	45
F	46
G	47
H	48
I	49
J	4A
K	4B

Name	Hex
L	4C
M	4D
N	4E
O	4F
P	50
Q	51
R	52
S	53
T	54
U	55
V	56

Name	Hex
W	57
X	58
Y	59
Z	5A

Answer:

Step 1: Compute RSA Keys

1.1 Calculate modulus n

$N = p \times q = 11 \times 13 = 143$

1.2 Compute $\phi(n)$

$\o(n) = (p - 1)(q - 1) = 10 \times 12 = 120$

1.3 Choose Public key E

E: $gcd(e, \phi(n)) = 1; 1 < E < \o(n)$

$gcd(E, 120) = 1$

$gcd(7, 120) = 1$

$E = 7$

1.4 Compute private key D

$(D \times E) \, mod \, ø(n) = 1$

$(D \times 7) \, mod \, 120 = 1$

By using Euclidem's algorithm

$D \times 7 \, mod \, 120 = 1$

$120 = n(7) + remainder$

$120 = 17(7) + 1$

Equation 1: $1 = 120 - 17(7)$

$1 = -17(7) + 120(1)$

$-17 \text{ is the modular inverse of } 7 \, mod \, 120$

Negative values cannot be directly utilized as RSA keys. Therefore, if the computed modular inverse D is negative, it must be converted to its positive equivalent within the modulus. This can be achieved by applying the transformation:

$D = ø(n) - |d|$

$D = 120 - 17 = 103$

$(103 \times 7) \, mod \, 120 = 1$

$D = 103$

Step 2: Convert "JEFFREY ULLMAN" to ASCII

Character	Hex	ASCII
J	4A	$(4 \times 16^{1}) + (10 \times 8^{0})$ $= 64 + 10$ $= 74$
E	45	$(4 \times 16^{1}) + (5 \times 8^{0})$ $= 64 + 5$ $= 69$
F	46	$(4 \times 16^{1}) + (6 \times 8^{0})$ $= 64 + 6$

		$= 70$
F	46	$(4 \times 16^1) + (6 \times 8^0)$ $= 64 + 6$ $= 70$
R	52	$(5 \times 16^1) + (2 \times 8^0)$ $= 80 + 2$ $= 82$
E	45	$(4 \times 16^1) + (5 \times 8^0)$ $= 64 + 5$ $= 69$
Y	59	$(5 \times 16^1) + (9 \times 8^0)$ $= 80 + 9$ $= 89$
(space)	20	$(2 \times 16^1) + (0 \times 8^0)$ $= 32 + 0$ $= 32$
U	55	$(5 \times 16^1) + (5 \times 8^0)$ $= 80 + 5$ $= 85$
L	4C	$(4 \times 16^1) + (12 \times 8^0)$ $= 64 + 12$ $= 76$
L	4C	$(4 \times 16^1) + (12 \times 8^0)$ $= 64 + 12$ $= 76$

M	4D	$(4 \times 16^1) + (13 \times 8^0)$ $= 64 + 13$ $= 77$
A	41	$(4 \times 16^1) + (1 \times 8^0)$ $= 64 + 1$ $= 65$
N	4E	$(4 \times 16^1) + (E \times 8^0)$ $= 64 + 14$ $= 78$

The plaintext in numeric form: [74, 69, 70, 70, 82, 69, 89, 32, 85, 76, 76, 77, 65, 78]

Step 3: Encrypt Each ASCII Using RSA Encryption

$CT = PT^E \ mod\ N$

Char	ASCII	P^7	$CT = PT^7 \ mod\ 143$
J	74	1.2151×10^{13}	35
E	69	7.4464×10^{12}	108
F	70	8.2354×10^{12}	60
F	70	8.2354×10^{12}	60
R	82	2.4929×10^{13}	69
E	69	7.4464×10^{12}	108
Y	89	4.4231×10^{13}	67
(space)	32	3.4360×10^{10}	98

U	85	3.2058×10^{13}	123
L	76	1.4645×10^{13}	54
L	76	1.4645×10^{13}	54
M	77	1.6049×10^{13}	77
A	65	4.9022×10^{12}	65
N	78	1.7566×10^{13}	78

Encrypted values: [35, 108, 60, 60, 69, 108, 67, 98, 123, 54, 54, 77, 65, 78]

Step 4: Decrypt Each Ciphertext Using RSA Decryption

$PT = CT^{D} \ mod\ N$

CT	C^{103}	$PT = C^{103} \ mod\ 14$	Character
35	1.0940×10^{159}	74	J
108	2.7710×10^{209}	69	E
60	1.4111×10^{183}	70	F
60	1.4111×10^{183}	70	F
69	2.5203×10^{183}	82	R
108	2.7710×10^{209}	69	E
67	1.2182×10^{188}	89	Y
98	1.2482×10^{205}	32	(space)
123	1.8207×10^{215}	85	U

54	2.7325×10^{178}	76	L
54	2.7325×10^{178}	76	L
77	2.0349×10^{194}	77	M
65	5.3713×10^{186}	65	A
78	7.6868×10^{194}	78	N

Decrypted numeric: [74, 69, 70, 70, 82, 69, 89, 32, 85, 76, 76, 77, 65, 78]

Decrypted text: "JEFFREY ULLMAN"

The Python built-in function pow(base, exponent, modulus) was utilized for encryption and decryption in the RSA algorithm in this question to ensure accuracy to avert overflow and maintain precision when handling large powers.

```
#J
C = pow(74, 7, 143)
print("Encrypted value (C):", C)

#J
D = pow(35, 103, 143)
print("Decrypted value (D):", D)
```

```
Encrypted value (C): 35
Decrypted value (D): 74
```

Figure 1.0

4

Question 5

- **Do a desktop research and write an essay for about five to six pages on the Public Awareness, Perception and Acceptance of Digital Signatures in general and in Malaysia. Identify the popular Certification Authorities (CAs) for Digital Signature in Malaysia.**

Public Awareness, Perception, and Acceptance of Digital Signatures in Malaysia

Abstract

Digital signatures are important for Malaysia's digital development. But many people still do not know about them or want to use them. This study looks at what Malaysian people know about digital signatures. It also looks at how they feel about using them. Research shows big differences between city people (75.6%) and village people (24.4%) who use the internet. This affects how much they know about digital signatures. Malaysia has good laws from 1997 and 2006. But many people do not understand the difference between electronic and digital signatures. Main problems include safety worries and not knowing how to use technology. Poor internet connections are also a problem, especially in villages and among older people. Malaysia wants everyone to use digital services by 2025. So it needs to fix these problems while using existing certification companies.

1. Introduction

1.1 Understanding Digital Signatures

Digital signatures are a basic part of modern digital checking in Malaysia. Electronic signatures are defined in the Electronic Commerce Act 2006. They include "any letter, character, number, sound or any other symbol created in electronic form adopted by a person as a signature" (Edwin Lee & Partners, 2024). Digital signatures are controlled by the Digital Signature Act 1997. They use special computer systems that provide safety checks (Zoho Sign, 2024). Digital signatures use public key systems. These systems make sure documents are real and complete. They also make sure people cannot deny they signed them.

1.2 Malaysia's Legal Framework

Malaysia's digital signature system has three main laws. These are the Digital Signature Act 1997, Electronic Commerce Act 2006, and Electronic Government Activities Act 2007 (Digital Policy Alert, 2024). The Malaysian Communications and Multimedia Commission watches over four licensed certification companies. These are MSC Trustgate, MyTrustID, SIRIM QAS International, and Cybersign Malaysia (MSC Trustgate, 2024).

2. Awareness - Public Knowledge and Understanding

2.1 Current State of Digital Signature Awareness

Malaysian people's knowledge about digital signatures shows wider patterns of digital knowledge. Research on Malaysia's National Digital Identity program shows that people's awareness was low. But their feelings and acceptance was quite good (ResearchGate, 2024). The Malaysian government's MyDigital ID program got over one million users in its first year. This shows people are becoming more familiar with digital identity ideas (Biometric Update, 2024). But many users still do not understand the difference between general digital identity and specific digital signature technologies.

2.2 Different Groups and Geographic Areas

Awareness is very different across different groups of people. Education, location, and age are the main factors. Education level is a strong predictor. University graduates show much higher awareness. The city-village divide is the biggest challenge. City areas have 75.6% internet users while village areas have only 24.4% (TM R&D, 2024). The Digital Competency Score 2023 report shows village school students scored 3.24 out of 5.00. The national average was 3.31 (Malay Mail, 2024).

2.3 Professional and Work Area Awareness

Government workers using the ePerolehan system show high awareness. This is because they must use it. Legal professionals also show high awareness through court e-filing systems. Small and medium businesses show different levels of awareness. This depends on how much they use digital technology. Consumer awareness stays limited even though the legal framework recognizes it (Adobe, 2024).

3. Perception - Public Attitudes and Trust

3.1 Trust and Security Views

People's thoughts about digital signature safety involve both technology understanding and cultural factors. Research shows privacy and safety concerns are big barriers to using them (ResearchGate, 2024). Younger people who know technology tend to view digital signatures positively. They recognize better safety features. Older people often doubt them. They prefer familiar paper-based processes. Cultural factors that stress face-to-face interactions create resistance to digital alternatives.

3.2 How Convenient and Legally Valid People Think They Are

How convenient people think digital signatures are depends on user experience and technology familiarity. Users who have successfully used them report big time savings. They also report better efficiency. But thinking they are complex remains a barrier. Certificate requirements make people think digital signatures are complex and take too much time. Many people do not understand that digital signatures are legally valid. This is even though there are good legal frameworks. Many Malaysians do not know that digital signatures are legally binding like handwritten signatures (Tay & Partners, 2021).

3.3 Cultural and Age Group Attitudes

Cultural attitudes show wider technology adoption patterns in Malaysia's multicultural society. Age differences are clear. Young people show positive attitudes toward digital signatures. Older generations stay more attached to traditional signing ceremonies and physical documentation. Media coverage of cybersecurity incidents greatly affects how people think. MyCERT reported 8,366 cybercrime cases in 2020. This adds to digital safety concerns (ResearchGate, 2024).

4. Section 3: Acceptance of Digital Signatures in Malaysia

Digital signatures have picked up momentum in Malaysia in recent years due to the New Crown pandemic that has turned online business into a sought-after phenomenon. Various organizations have started embracing digital signature technology for workflow improvement and document authenticity.

Universiti Teknologi and Universiti Malaya are among the universities that have begun issuing digitally signed e-transcripts and graduation certificates. These are able to successfully prevent forgery and speed up the verification process when looking for work or furthering studies abroad.

Digital signatures have also been introduced in certain government agencies for specific services. As an example, Malaysia's Inland Revenue Department (LHDN) has included digital signatures in formal documents such as electronic returns of taxation and assessment notice in an effort to enable taxpayers and third-party auditors to authenticate and ensure integrity of documents (INLAND REVENUE BOARD OF MALAYSIA, 2025).

The promotion of digital signatures is also supported by law. Under the Digital Signature Act 1997, digital signatures are legally valid and the government has explicitly authorized Certification Authorities (CAs) to assume responsibility for verification to ensure the legitimacy and security of the signing process (Lee, Perara & Tan, 2022).

Despite the gradual expansion of adoption, there is still a gap in overall public adoption. According to a survey conducted by the Malaysian Digital Economy Corporation (MDEC) in 2023, more than 65% of medium and large enterprises have implemented some kind of digital signing solution, but only 28% of SMEs and 15% of general users have reported practical experience with it, with the main barriers being lack of awareness, lack of trust, and complexity of the technology process (*Malaysia Digital*, n.d.).

With government-led initiatives such as the MyDigital ID digital identity system and the National Digital Identity Framework (NDIF) continuing to advance, the acceptance of digital signatures in Malaysia is expected to grow steadily in the coming years.

5. Section 4: Factors Influencing the Adoption of Digital Signatures in Malaysia

Despite the upward trend in the use of digital signatures in Malaysia, there are still significant differences in adoption across different fields and user groups. This imbalance is analyzed in terms of the key factors driving and hindering adoption.

5.1 Enablers: Legal Security and Institutional Trust

Malaysia established the legal basis for digital signatures as early as 1997.The Digital Signature Act 1997 clearly gives legal effect to digital signatures, giving businesses and government agencies legal confidence in the use of digital signatures (Lee, Perara & Tan, 2022). In addition, for example, the Malaysian Inland Revenue Department (LHDN) has to included digital signatures in its electronic invoices, which not only ensures the authenticity of the documents, but also serves as an example of the government's pioneering adoption of digital signatures (LHDN, 2024), which further enhances the public's trust.

5.2 Government Infrastructure Driving Adoption

MyDigital ID and the National Digital Identity Framework (NDIF) launched by the government are dedicated to unifying the digital authentication process, laying the foundation for reducing the reliance on physical signatures and promoting a digital trust environment.

5.3 Impediments: Misperceptions and Trust Gaps

Although the legal framework is in place, the public's understanding of digital signature technology is still quite limited. Many people still mistake “e-signature” for a simple image of a signature font rather than an authentic verification mechanism based on cryptography. According to a report by the Malaysian Digital Economy Corporation (MDEC) in 2023, only 28% of SMEs and 15% of general users actually use digital signatures, reflecting that barriers to use and trust issues still exist (MDEC, 2023).

Integrating Certification Authorities (CAs) and maintaining systems are often expensive or difficult technical challenges for small businesses to master. The lack of easy-to-use tools and clear guidelines also makes adoption more costly than beneficial. Overall, Malaysia has a well-established legal and institutional foundation, but it is important to further educate the public, reduce the cost of authentication, and increase ease of use in order to truly popularize digital signatures at all levels.

6. Certification Authorities for Digital Signatures in Malaysia

Under the Digital Signature Act 1997, the use of digital signatures must be certified by a Certification Authority (CA) authorized by the Malaysian Communications and Multimedia Commission (MCMC) in order to be legally valid. The following are the four local CAs that are currently officially recognized:

Certification Authority	**Description**	**Official Website**
MSC Trustgate Sdn Bhd	Authorized by MCMC since 2000, providing e-signature, PKI digital certificates, time-stamping and document sealing services, widely used in government, enterprises and financial institutions.	https://www.msctrustgate.com
Pos Digicert Sdn Bhd	A Pos Malaysia organization specializing in digital certificate services in the financial, legal, government and e-government sectors.	https://www.posdigicert.com.my
Telekom Applied Business (TM SIGN)	Part of Telekom Malaysia, providing digital signature solutions that can be integrated into telecom and enterprise systems.	http://www.tmca.com.my
Raffcomm Technologies Sdn Bhd	Provide secure email, document signing and authentication services based on public key infrastructure (PKI)	https://www.rafftech.my

These certification authorities are responsible for issuing, validating and managing legitimate digital signature certificates to ensure that they are legally recognizable for a wide range of cross-industry e-contracting, tax, banking, university and government platforms.

Different Certification Authorities have their own specialties, such as Pos Digicert, which is commonly used for contract signing in the legal industry, and TM SIGN, which is more suited to the integration needs of large corporations and telecommunication systems. The diversity of these organizations creates a robust and trusted framework for digital signature services in Malaysia.

7.0 Conclusion

Digital signatures in Malaysia depend on three things. These are laws, technology working, and how much people know about them. The Digital Signature Act 1997, the

E-Commerce Act 2006, and new computer security rules give strong legal help for digital signatures. These legislations also entail elaborate guidelines for their enforcement. Digital signatures, on the other hand, are enforced uniformly throughout the nation by individuals.This study confirmed that they are not yet adequately educated regarding digital signatures. The situation is further critical in rural areas and with the elderly. Some groups have started using digital signatures in their systems. These include the Inland Revenue Department of Malaysia (LHDN) and the University of Malaya (UM). But many small businesses and regular users are still scared to use digital signatures. They have technology problems and do not have clear help. People also mix up "electronic signatures" and "digital signatures with secret codes." This mix-up has made people trust these tools less.

But there are good signs for the future. The government made MyDigital ID and the National Digital Identity Framework (NDIF). These programs are slowly helping people learn about digital technology. They are also building better systems across the country. Allowed companies that check signatures are also helping. These companies include MSC Trustgate, Pos Digicert, TM SIGN and Raffcomm Technologies. They are building safe and trusted places for digital signature services.

5.5/10

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